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RAD: Rule-Augmented Relational Anomaly Detection

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PAPER
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CODE & DATA
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01 Anomalies live in relations, not rows

In cybersecurity, fraud, and e-commerce the data is a **multi-table relational database**: users, machines, products, and the events that link them. Most anomaly detectors still require one flat feature matrix, so practitioners join and aggregate the tables into row-level summaries.

Flattening discards exactly what makes many anomalies anomalous: **entity identity, schema roles, multi-hop dependencies, and temporal context**.

> *An authentication event can look unremarkable on its own yet be anomalous because* **this user–machine pair never co-occurred**, *the machine is otherwise touched only by service accounts, and the user authenticated to four other machines in the previous ten minutes. Recovering those facts needs joins across history and neighbors.*

Relational deep learning preserves structure but has no anomaly objective. Symbolic behavioral rules capture analyst-style evidence but are typically bolted on after scoring. **RAD puts symbolic evidence inside the learned relational representation.**

02 Problem: relational anomaly detection

Given a database $\mathcal{D}$ with schema $\mathcal{S}$, a target table T_{target}, its heterogeneous graph $\mathcal{G} = (V, E, \tau)$, and labels on a training subset $V_{\text{train}} \subset V_{\text{target}}$, learn a scoring function

$$s : V_{\text{target}} \to \mathbb{R} \quad \text{with} \quad s(v_i) > s(v_j) \ \text{ whenever } \ y_i = 1, y_j = 0$$

evaluated by **ranking** held-out nodes V_{test} (disjoint from training). With timestamps, $s(v)$ may only use structure and attributes at or before t_v, so no future information leaks into a score. The goal is a ranking under severe class imbalance, not a calibrated label.

03 Contributions

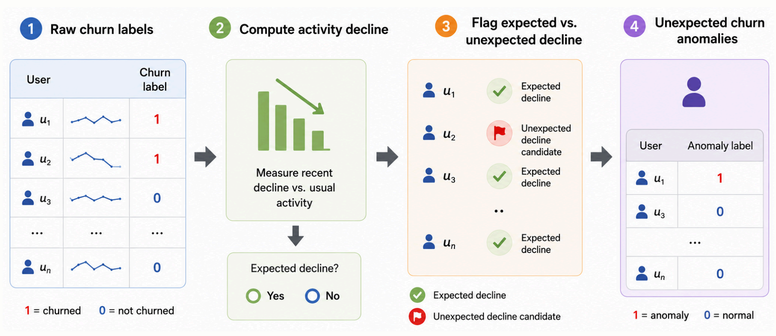
- **Formalize** relational anomaly detection as distinct from tabular and graph anomaly detection: anomalousness can depend on cross-table, multi-hop, temporal, and schema-dependent interactions.
- **RAD**, a detector that mines rules from a flattened view, injects them into target nodes *before* message passing, and trains a heterogeneous graph autoencoder with reconstruction and pairwise-ranking objectives.
- **A benchmark** of three tasks (LANL cybersecurity, Amazon and H&M unexpected churn) with unified target construction and a ranking-based protocol.
- **Evidence** that direct rule injection and ranking supervision drive the gains, while edge reconstruction is not uniformly beneficial.

04 Benchmark

Task	Target	Tables	Records	Targets	Anomaly
LANL	Auth event	7	1.65B	1.64B	Red-team authentication
Amazon	Customer	3	15.0M	1.85M	Unexpected review churn
H&M	Customer	3	16.7M	1.37M	Unexpected purchase churn

LANL scores authentication events in sampled relational subgraphs (users, source and destination computers, process, DNS, and flow records) with red-team labels as ground truth.

Unexpected churn (Amazon, H&M; RelBench databases) treats churn as anomalous only when it is *not* preceded by an expected activity decline. A churned user u is removed when its recent activity change $g_u \le -\theta(\mu_u)$, with a threshold that adapts to the baseline activity level μ_u. Default prevalence 1.5%; a sweep over 0.5%, 1.5%, and 3.0% tests sensitivity to this choice.



Unexpected-churn filtering: churned users with an expected prior decline are dropped; the rest become anomalies.

05 What a mined rule looks like

From the released LANL rule files. A random-forest path is a long conjunction of split thresholds; refinement keeps the behavioral core.

RANDOM-FOREST PATH

```
auth_type > 4.5 ∧ auth_type ≤ 5.5 ∧ new_src_dst_user_pair
> 0.5 ∧ user_auth_count_24h ≤ 0.5 ∧ success_failure > 0.5
```

5 conjuncts; sampled paths in this run carry 5–19 conjuncts each.

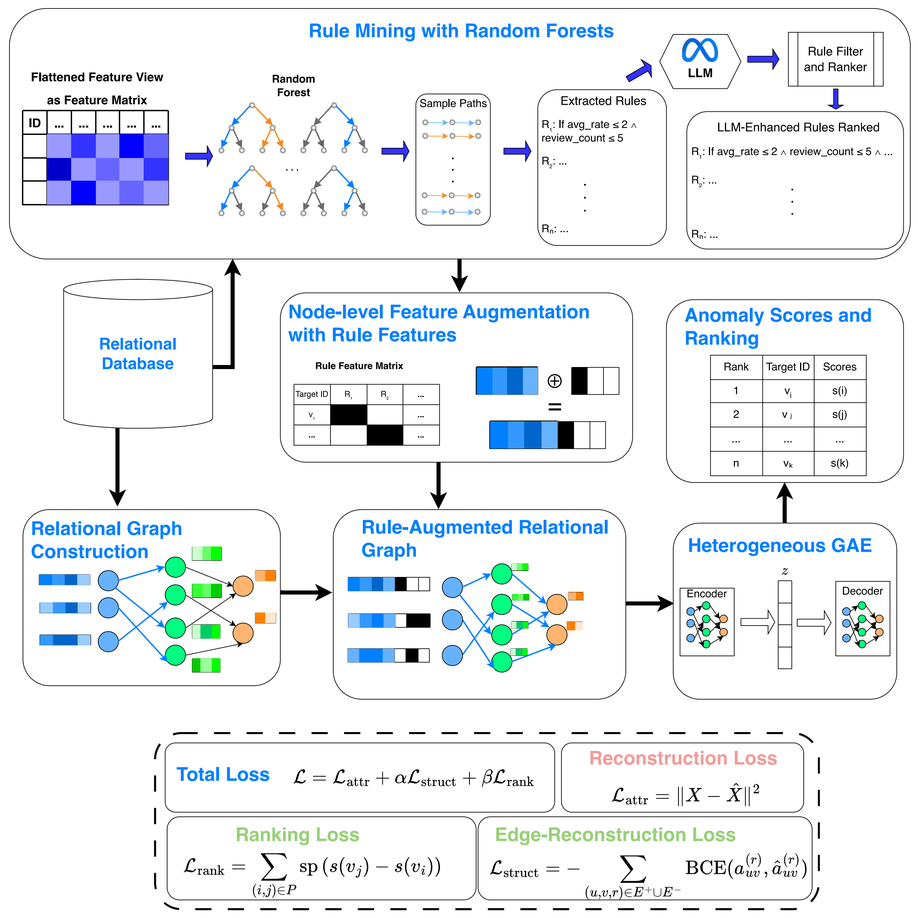
↓ LLM proposes · same support / confidence / lift filter re-validates

COMPACT PREDICATE

```
new_src_dst_user_pair = 1 ∧ user_auth_count_24h > 10
```

A never-before-seen user → machine pair from a user with more than ten authentications in the past 24 hours. Each such predicate becomes one binary feature on the target node.

06 RAD in four stages



A relational database is viewed two ways: a flattened view for rule mining and a heterogeneous graph for representation learning. Random-forest paths become predicates, predicates become binary node features, and a heterogeneous GAE scores and ranks target nodes.

1 Relational graph construction

Rows become typed nodes, primary–foreign keys become typed edges. Type-specific input projections handle different feature spaces; construction respects the temporal constraint.

2 Rule mining & validation

A random forest is trained on an auxiliary flattened view of the target entities. Each root-to-leaf path is a conjunctive predicate $q(v) = c_1 \wedge \cdots \wedge c_m$. Predicates touching identifiers, labels, or future columns are dropped; the rest are ranked by **support, confidence, and lift** computed on training/validation data only.

Optionally, an LLM proposes SQL `WHERE` predicates that prune, tighten, or combine seed rules. Every proposal is re-executed and re-scored by the same support/confidence/lift filter; the LLM never sees test labels and never assigns scores. The rule set is frozen before testing.

3 Rule-augmented node features

Each retained rule is a binary feature; the rule vector $r(v) \in \{0,1\}^K$ is concatenated onto the target node: $\tilde{x}_v = [x_v \,\|\, r(v)]$. Non-target nodes keep their original features.

4 Heterogeneous GAE & ranking objective

A two-layer relation-aware GraphSAGE encoder (mean aggregation, hidden size 128) yields z_v; a decoder reconstructs $\tilde{x}_v$. The anomaly score is the reconstruction error, optionally combined with an edge-reconstruction score.

$$\mathcal{L} = \mathcal{L}_{\text{attr}} + \alpha\,\mathcal{L}_{\text{struct}} + \beta\,\mathcal{L}_{\text{rank}}, \quad \mathcal{L}_{\text{rank}} = \mathbb{E}_{(i,j)\sim P\times N}\,\text{softplus}(s_j - s_i)$$

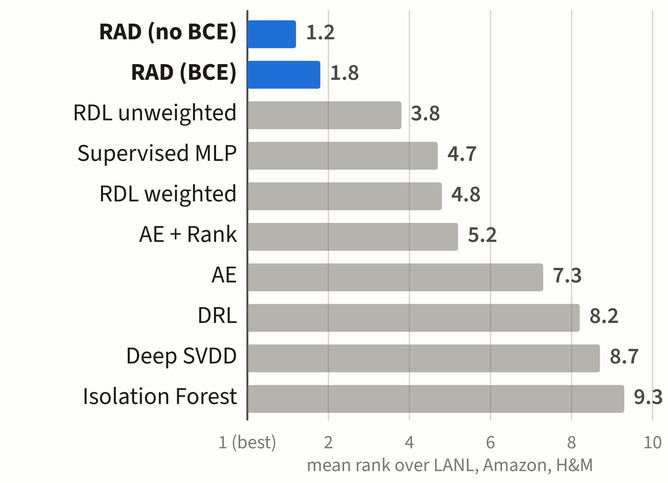
Design choice: rules enter *before* message passing, so symbolic evidence shapes the learned embeddings and interacts with relational context, rather than adjusting scores after the fact (post-hoc fusion).

07 Results: 10 seeds, mean ± std, best per column in bold

	LANL			Amazon			H&M			Avg. rank ↓		
Model	**AUROC**	**AUPRC**	**P@k**	**AUROC**	**AUPRC**	**P@k**	**AUROC**	**AUPRC**	**P@k**	**AUROC**	**AUPRC**	**P@k**
RAD (rules, no BCE)	**0.996**$_{\pm.001}$	**0.659**$_{\pm.053}$	**0.601**$_{\pm.012}$	0.693$_{\pm.010}$	**0.043**$_{\pm.003}$	0.050$_{\pm.007}$	**0.828**$_{\pm.071}$	**0.143**$_{\pm.017}$	**0.129**$_{\pm.022}$	**2.0**	**1.2**	**2.0**
RAD (rules, BCE)	0.860$_{\pm.155}$	0.557$_{\pm.029}$	0.572$_{\pm.008}$	0.695$_{\pm.010}$	**0.043**$_{\pm.004}$	0.051$_{\pm.004}$	0.787$_{\pm.074}$	0.110$_{\pm.020}$	0.125$_{\pm.022}$	3.3	1.8	2.3
RDL weighted [Fey et al.]	0.553$_{\pm.158}$	0.007$_{\pm.014}$	0.021$_{\pm.064}$	**0.720**$_{\pm.002}$	0.037$_{\pm.000}$	**0.062**$_{\pm.004}$	0.700$_{\pm.005}$	0.035$_{\pm.001}$	0.064$_{\pm.013}$	4.7	4.8	3.7
RDL unweighted [Fey et al.]	0.685$_{\pm.205}$	0.015$_{\pm.028}$	0.026$_{\pm.078}$	0.713$_{\pm.002}$	0.037$_{\pm.001}$	**0.062**$_{\pm.005}$	0.693$_{\pm.019}$	0.039$_{\pm.004}$	0.064$_{\pm.016}$	4.3	3.8	3.3
AE + Rank	0.943$_{\pm.035}$	0.525$_{\pm.078}$	0.540$_{\pm.065}$	0.518$_{\pm.014}$	0.016$_{\pm.001}$	0.015$_{\pm.003}$	0.456$_{\pm.024}$	0.013$_{\pm.000}$	0.005$_{\pm.004}$	5.2	5.2	5.7
DRL [Ye et al.]	0.673$_{\pm.159}$	0.004$_{\pm.002}$	0.005$_{\pm.004}$	0.463$_{\pm.022}$	0.013$_{\pm.001}$	0.003$_{\pm.000}$	0.456$_{\pm.012}$	0.013$_{\pm.001}$	0.009$_{\pm.009}$	7.5	8.2	7.2
AE	0.888$_{\pm.046}$	0.009$_{\pm.003}$	0.000$_{\pm.000}$	0.474$_{\pm.013}$	0.014$_{\pm.001}$	0.014$_{\pm.009}$	0.408$_{\pm.020}$	0.012$_{\pm.000}$	0.007$_{\pm.005}$	6.7	7.3	7.8
Deep SVDD	0.736$_{\pm.073}$	0.005$_{\pm.002}$	0.001$_{\pm.003}$	0.458$_{\pm.031}$	0.013$_{\pm.001}$	0.003$_{\pm.001}$	0.412$_{\pm.023}$	0.012$_{\pm.001}$	0.002$_{\pm.003}$	7.7	8.7	8.8
Isolation Forest	0.644$_{\pm.018}$	0.003$_{\pm.000}$	0.000$_{\pm.000}$	0.450$_{\pm.002}$	0.013$_{\pm.000}$	0.001$_{\pm.000}$	0.402$_{\pm.014}$	0.012$_{\pm.000}$	0.004$_{\pm.005}$	9.7	9.3	9.5
Supervised MLP (full labels)	0.965$_{\pm.003}$	0.050$_{\pm.011}$	0.093$_{\pm.035}$	0.659$_{\pm.024}$	0.027$_{\pm.003}$	0.041$_{\pm.009}$	0.642$_{\pm.028}$	0.027$_{\pm.005}$	0.039$_{\pm.023}$	4.0	4.7	4.7

RDL = relational deep learning GNN on the same heterogeneous graph without rules. AE, Deep SVDD, Isolation Forest, DRL operate on the flattened view. P@k uses k = number of positives in the test split. Avg. rank is per metric across the three datasets (lower is better).

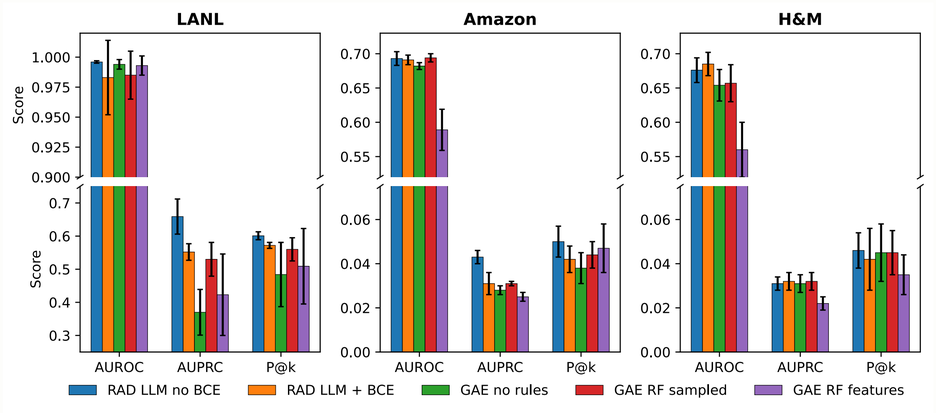
Average AUPRC rank across datasets



RQ1. RAD has the best average AUPRC rank. On LANL it dominates every baseline on AUPRC and P@k. On Amazon, supervised RDL wins AUROC and P@k while RAD wins AUPRC; on H&M, RAD wins all three metrics.

RQ5. Sweeping the unexpected-churn prevalence over 0.5%, 1.5%, and 3.0% leaves the picture intact: RAD stays competitive with or ahead of supervised RDL on AUPRC at every level, so the conclusions do not hinge on one labeling threshold.

Ablations: rule construction and edge reconstruction (10 seeds)



Broken y-axis separates AUROC from the much smaller AUPRC and P@k. **RQ2:** rule-augmented variants beat the no-rule GAE on AUPRC and P@k; compact predicates beat raw random-forest features. **RQ3:** dropping edge reconstruction (no BCE) is clearly better on LANL and H&M and tied on Amazon. **RQ4:** LLM refinement helps most on LANL; elsewhere it acts as a rule-compaction step rather than a new signal.

08 Takeaways

- Relational anomaly detection is a joint problem of **representation, scoring, and behavioral evidence**; solving one piece is not enough.
- Mined rules are most useful when **injected into the graph before message passing**, not fused with scores afterwards.
- **Pairwise ranking supervision** is what turns a reconstruction model into a strong retriever of rare anomalies.
- Edge reconstruction is an **optional auxiliary**: it can overemphasize sampled topology that is only loosely tied to the anomaly label.

09 Limitations & next steps

- Rule mining is bounded by the flattened view and needs labels for the random forest and the support/confidence/lift ranking; RAD does not yet apply to unlabeled settings.
- Amazon and H&M results use a held-out partition of the validation split, since RelBench withholds test labels; LANL uses sampled subgraphs around anchor events.
- Next: unsupervised rule discovery, rules defined over multi-hop relational structure rather than flattened summaries, richer relational backbones, and rules as human-readable explanations for flagged anomalies.

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